

1	Course Name:
	Network Fundamentals
2	Course Code:
	INB12604
3	Year and Semester Offered:
	Year 1 Semester 2
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NIL
6	Course Synopsis:
	This course introduces the architecture, structure, functions, components, and models of the Internet and other computer networks. The principles and structure of IP addressing and the fundamentals of Ethernet concepts, media, and operations are introduced to provide a foundation for the curriculum. By the end of the course, students will be able to build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Explain the communication processes and protocols in networking 2. Display the ability to configure simple networks 3. Prepare proposals for small network designs based on user requirements
8	Content Outline:
	Topic
	1. Introduction to Networking
	2. Basic Switch and End Device Configuration
	3. Protocols and Models
	4. Physical Layer
	- Transmission Media - Numbering System
	5. Data Link Layer
	- Ethernet
	6. Network Layer
	- IP Addressing - Basic Router Configuration - ICMP
	7. Transport Layer
	8. Application Layer

	9. Network Security Fundamentals
	10. Build a Small Network - Network Troubleshooting
	11. Bridging Topics